# Math 110 — Week 6

## Invariant subspaces, eigenvalues and eigenvectors, PageRank (Notes Ch. 11, LADR 5A, HW6)

## §1 Facts used from Weeks 1–5

(W1) In a space of dimension $n$, any $n+1$ vectors are dependent, and an independent list has at most $n$ vectors (Week 2).

(W2) Dependence lemma: in a dependent list $v_1, \dots, v_m$ some $v_k$ lies in $\operatorname{span}(v_1, \dots, v_{k-1})$ (Week 2).

(W3) For an operator $T$ on a finite-dimensional space: injective $\iff$ surjective $\iff$ invertible. $T$ is injective $\iff \operatorname{null} T = \{0\}$ (Weeks 3, 4).

(W4) $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is invertible $\iff \det A = ad - bc \neq 0$ (Week 4).

(W5) $\mathcal{M}(T)\mathcal{M}(v) = \mathcal{M}(Tv)$: applying $T$ multiplies the coordinate column by the matrix (Week 3).

(W6) A stochastic matrix $P$ has entries $\geq 0$ and each column sums to 1. If $p$ is a probability vector, so is $Pp$ (Week 3).

(W7) If $T(U_1) \subseteq U_2$, then $v + U_1 \mapsto Tv + U_2$ is a well-defined linear map $V/U_1 \to W/U_2$ (Week 5).

## §2 This week

### §2.1 Invariant subspaces, eigenvectors, eigenspaces (Ch. 11.1)

**The question.** An operator $T\colon V \to V$ maps a space to itself, so it can be applied repeatedly. To understand $T$ we try to cut $V$ into smaller pieces that $T$ maps into themselves, and study $T$ on each piece. What are the smallest such pieces?

**Definition** (Def. 30 / LADR 5.2)**.** Let $T \in \mathcal{L}(V)$. A subspace $U \subseteq V$ is **invariant under** $T$ if $Tu \in U$ for every $u \in U$.

- To check invariance it is enough to check $Tu \in U$ for $u$ in a spanning list of $U$.
- If $U$ is invariant, $T$ gives two smaller operators: the **restriction** $T|_U\colon U \to U$, and the **quotient operator** $T/U\colon V/U \to V/U$, $(T/U)(v+U) = Tv + U$ (well-defined by (W7) with $U_1 = U_2 = U$).
- Ex. 48: $\{0\}$, $V$, $\operatorname{null} T$ and $\operatorname{range} T$ are always invariant. ($u \in \operatorname{null} T \Rightarrow Tu = 0 \in \operatorname{null} T$. $u \in \operatorname{range} T \Rightarrow Tu \in \operatorname{range} T$ because $Tu$ is an output of $T$.)

**Definition** (Def. 31, 32 / LADR 5.5, 5.8, 5.52)**.** A *nonzero* vector $v$ is an **eigenvector** of $T$ if $Tv = \lambda v$ for some $\lambda \in \mathbb{F}$. The scalar $\lambda$ is the **eigenvalue** of $v$. The **eigenspace** of $\lambda$ is

$$E(\lambda, T) = \operatorname{null}(T - \lambda I) = \{v : Tv = \lambda v\}.$$

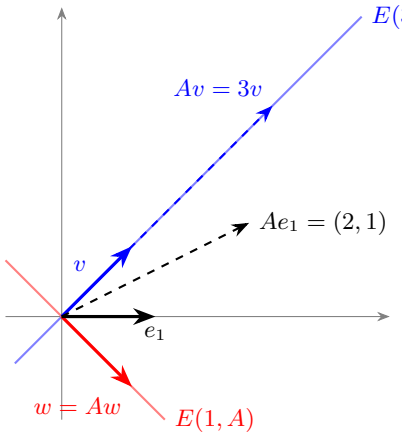
- **An eigenvector is the same as a one-dimensional invariant subspace.** If $\operatorname{span}(v)$ is invariant, then $Tv \in \operatorname{span}(v)$, so $Tv = \lambda v$. Conversely if $Tv = \lambda v$ then $T(cv) = c\lambda v \in \operatorname{span}(v)$.
- $\lambda$ is determined by $v$: if $\lambda v = \mu v$ with $v \neq 0$ then $\lambda = \mu$.
- $E(\lambda, T)$ consists of 0 and all eigenvectors with eigenvalue $\lambda$. It is a subspace (a null space), and it is invariant: if $Tv = \lambda v$ then $T(Tv) = \lambda(Tv)$.
- $\lambda$ is an eigenvalue of $T$ exactly when $E(\lambda, T) \neq \{0\}$.

**Proposition** (LADR 5.7: four ways to say "$\lambda$ is an eigenvalue")

Let $V$ be finite-dimensional. The following are equivalent: (a) $\lambda$ is an eigenvalue of $T$; (b) $T - \lambda I$ is not injective; (c) $T - \lambda I$ is not surjective; (d) $T - \lambda I$ is not invertible.

*Proof.* $Tv = \lambda v$ is the same equation as $(T - \lambda I)v = 0$. So (a) says $\operatorname{null}(T - \lambda I) \neq \{0\}$, which is (b). For an operator on a finite-dimensional space, (b), (c), (d) are equivalent (W3). □

In particular: 0 **is an eigenvalue of** $T \iff T$ **is not invertible**, and $E(0, T) = \operatorname{null} T$.




$A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$.
$v = (1, 1)$: $Av = (3, 3) = 3v$. The line through $v$ is mapped into itself.
$w = (1, -1)$: $Aw = (1, -1) = w$.
$e_1 = (1, 0)$: $Ae_1 = (2, 1)$ is not a multiple of $e_1$. The line through $e_1$ is not invariant.

Eigenvectors are the vectors whose direction is kept. Solid arrows are vectors, dashed arrows their images.

**Example** (Ex. 49: all invariant subspaces of three operators on $\mathbb{R}^2$)

In $\mathbb{R}^2$ the subspaces are $\{0\}$, lines, and $\mathbb{R}^2$. An invariant line is $\operatorname{span}(v)$ with $v$ an eigenvector. So finding invariant subspaces means finding eigenvectors.

1. $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$: $(x, y) \mapsto (x, 2y)$ is a multiple of $(x, y)$ only if $x = 0$ or $y = 0$. Invariant: $\{0\}$, $\operatorname{span}(e_1)$, $\operatorname{span}(e_2)$, $\mathbb{R}^2$.
2. $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (rotation by 90°): no nonzero vector is mapped to a multiple of itself. Invariant: only $\{0\}$ and $\mathbb{R}^2$.
3. $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: $N(x, y) = (y, 0) = \lambda(x, y)$ forces $\lambda y = 0$ and $y = \lambda x$, so $y = 0$ (and $\lambda = 0$). Invariant: $\{0\}$, $\operatorname{span}(e_1)$, $\mathbb{R}^2$. The line $\operatorname{span}(e_2)$ is not invariant because $Ne_2 = e_1$.

For $N$ the chain $\{0\} \subset \operatorname{span}(e_1) \subset \mathbb{R}^2$ consists of invariant subspaces. Chains of invariant subspaces $\operatorname{span}(v_1) \subset \operatorname{span}(v_1, v_2) \subset \dots$ are what Week 7 uses to bring a matrix to triangular form.

**Stationary distributions are eigenvectors.** For the stochastic matrix of Week 3,

$$P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix},$$

the probability vector $\pi = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ satisfies $P\pi = \pi$: it is an eigenvector with eigenvalue 1. A probability vector with $P\pi = \pi$ is called a **stationary distribution**; then $P^k\pi = \pi$ for all $k$. Four separate questions: does one exist, is it unique, does $P^k p_0$ converge to it, and how fast.

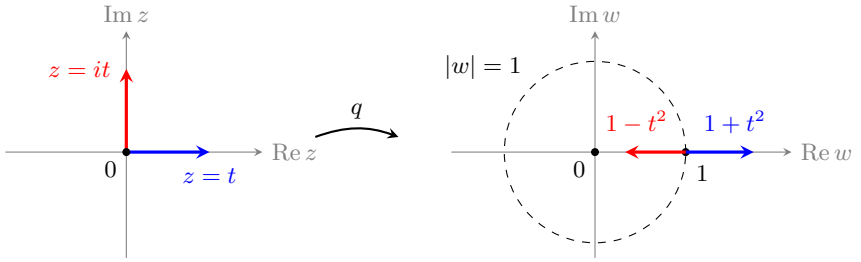
## §2.2 Complex numbers and the Fundamental Theorem of Algebra (Ch. 11.2)

**The question.** Does every operator have an eigenvector? Rotation of $\mathbb{R}^2$ by 90° has none. What changes over $\mathbb{C}$?

- For $z = a + bi$: $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$, the distance from $z$ to 0. $|zw| = |z||w|$ and $|z + w| \le |z| + |w|$.
- Polar form: $z = r(\cos\theta + i\sin\theta)$ with $r = |z|$. De Moivre: $z^k = r^k(\cos k\theta + i\sin k\theta)$. So every nonzero $z$ has a $k$-th root:

$$r^{1/k}\big(\cos\frac{\theta}{k} + i\sin\frac{\theta}{k}\big).$$

The key example: near $z = 0$, the polynomial $q(z) = 1 + z^2$ has $|q(0)| = 1$, and some directions make $|q|$ smaller.




inputs $z$ near 0 values $q(z)$ near $q(0) = 1$

For $q(z) = 1 + z^2$: inputs along the real axis move $q$ away from 0, outside the circle $|w| = 1$. Inputs along the imaginary axis have $z^2 = -t^2$, which points opposite to 1, so they move $q$ toward 0, inside the circle. So $|q|$ has no minimum at 0.

**Theorem** (Prop. 52: Fundamental Theorem of Algebra)
Every nonconstant polynomial with complex coefficients has a zero in $\mathbb{C}$.

**The idea**

Look at the size $|p(z)|$. Far from 0 it is large, so it has a smallest value somewhere, at a point $\zeta$. We want to show that smallest value is 0. Suppose it is not. Then we will find a nearby point where $|p|$ is even smaller, a contradiction.

Why can we always go lower from a nonzero value? Look at $q(z) = 1 + z^2$ at $z = 0$ (the figure above), where $|q| = 1$. Moving along the real axis, $q(t) = 1 + t^2$ gets bigger. Moving along the imaginary axis, $q(it) = 1 - t^2$ gets smaller. The term $z^2$ can be made to point in any direction by choosing the direction of $z$; choose it so that $z^2$ points opposite to the constant 1. In general, after moving $\zeta$ to the origin and dividing by $p(\zeta)$, the polynomial is $q(z) = 1 + a_k z^k +$ (higher powers) with $a_k \neq 0$. Choose the direction $\beta$ of $z$ so that $a_k\beta^k = -1$ (possible because $k$-th roots exist in $\mathbb{C}$). Along $z = t\beta$ the polynomial is $1 - t^k +$ (terms of size about $t^{k+1}$), and for small $t$ the term $-t^k$ wins.

*Proof.* Let $p(z) = c_0 + c_1 z + \cdots + c_m z^m$ with $m \ge 1$, $c_m \neq 0$.

1. $|p(z)|$ *is large far away.* By the triangle inequality $|p(z)| \ge |c_m||z|^m - \sum_{j<m} |c_j||z|^j$, and the right side tends to $\infty$ as $|z| \to \infty$. So there is a radius $R$ with $|p(z)| > |p(0)|$ whenever $|z| > R$.
2. *A global minimum exists.* On the closed disk $|z| \le R$ the continuous real function $|p|$ attains a minimum, at some $\zeta$. That minimum is $\le |p(0)|$, and outside the disk $|p|$ is larger than $|p(0)|$. So $|p(\zeta)| \le |p(z)|$ for all $z \in \mathbb{C}$.
3. *Suppose* $p(\zeta) \neq 0$. Let $q(z) = p(\zeta + z)/p(\zeta)$. Then $q(0) = 1$ and $|q(z)| \ge 1$ for all $z$.
4. $q$ is a nonconstant polynomial, so $q(z) = 1 + a_k z^k + a_{k+1} z^{k+1} + \cdots + a_m z^m$ where $a_k z^k$ is the first nonzero term after the constant.
5. Choose $\beta \in \mathbb{C}$ with $\beta^k = -1/a_k$. For real $t > 0$: $q(t\beta) = 1 - t^k + a_{k+1}\beta^{k+1}t^{k+1} + \cdots + a_m\beta^m t^m$.

6. For $0 < t < 1$ the tail has absolute value at most $Ct^{k+1}$, where $C = \sum_{j>k} |a_j \beta^j|$. So

$$|q(t\beta)| \leq (1 - t^k) + Ct^{k+1} = 1 - t^k(1 - Ct).$$

7. For $t$ small enough that $Ct < 1$, this is $< 1$. That contradicts $|q| \geq 1$. Hence $p(\zeta) = 0$.

□

- **Factorisation.** If $p(\lambda) = 0$ then $p(z) = (z - \lambda)q(z)$ with $\deg q = \deg p - 1$. Applying the theorem repeatedly: $p(z) = c(z - \lambda_1) \cdots (z - \lambda_m)$, roots listed with multiplicity. $\mathbb{C}$ is called **algebraically closed**.
- $\mathbb{R}$ is not: $x^2 + 1$ has no real zero.
- The only analytic input is that a continuous function on a closed bounded set attains its minimum.

## §2.3 Existence and independence of eigenvectors (Ch. 11.2)

**Theorem** (Prop. 53 / LADR 5.19: operators on complex spaces have eigenvectors)
Every operator on a nonzero finite-dimensional complex vector space has an eigenvector.

**The idea**

We need a polynomial so that the Fundamental Theorem can be used, but so far $T$ has no polynomial attached to it. Finite dimension produces one. Take any $v \neq 0$ and keep applying $T$: the vectors $v, Tv, T^2v, \ldots, T^nv$ are $n + 1$ vectors in a space of dimension $n$, so they satisfy a relation $a_0v + a_1Tv + \cdots + a_mT^mv = 0$. Read that as "$p(T)v = 0$" for the polynomial $p(z) = a_0 + a_1z + \cdots + a_mz^m$.

Now factor: $p(z) = (z - \lambda)q(z)$. Then $(T - \lambda I)\big(q(T)v\big) = p(T)v = 0$. So $w = q(T)v$ satisfies $Tw = \lambda w$. It is an eigenvector provided $w \neq 0$. To guarantee that, take the relation of *smallest* degree at the start: $q$ has smaller degree than $p$, so $q(T)v = 0$ would be a shorter relation.

Example: $T$ = rotation by 90° on $\mathbb{C}^2$, $v = e_1$. $Te_1 = e_2$, $T^2e_1 = -e_1$, so $T^2v + v = 0$: $p(z) = z^2 + 1 = (z - i)(z + i)$. Take $\lambda = i$, $q(z) = z + i$: $w = Te_1 + ie_1 = (i, 1)$, and $Tw = (-1, i) = i \cdot (i, 1)$.

*Proof.* Let $n = \dim V$ and choose $v \neq 0$.

1. $v, Tv, \ldots, T^nv$ are $n + 1$ vectors, so they are dependent (W1). Among all nontrivial relations $a_0v + a_1Tv + \cdots + a_mT^mv = 0$ ($a_m \neq 0$) choose one with $m$ as small as possible. Then $m \geq 1$, because $a_0v = 0$ with $a_0 \neq 0$ is impossible.
2. Let $p(z) = a_0 + a_1z + \cdots + a_mz^m$. It has a zero $\lambda \in \mathbb{C}$, so $p(z) = (z - \lambda)q(z)$ with $q(z) = b_0 + \cdots + b_{m-1}z^{m-1}$, $b_{m-1} = a_m \neq 0$.
3. Let $w = b_0v + b_1Tv + \cdots + b_{m-1}T^{m-1}v$. Then $w \neq 0$: otherwise this would be a nontrivial relation of degree $m - 1 < m$.
4. Multiplying out $(z - \lambda)q(z) = p(z)$ with $T$ in place of $z$:

$$(T - \lambda I)w = a_0v + a_1Tv + \cdots + a_mT^mv = 0.$$

5. So $Tw = \lambda w$ with $w \neq 0$.

□

- Complex is needed: rotation by 90° on $\mathbb{R}^2$. Finite-dimensional is needed: $T(p) = xp$ on $\mathcal{P}(\mathbb{C})$ has no eigenvector, since $xp = \lambda p$ fails by comparing degrees.

**Theorem** (Prop. 54 / LADR 5.11: eigenvectors for distinct eigenvalues are independent)
If $v_1, \dots, v_m$ are eigenvectors of $T$ with distinct eigenvalues $\lambda_1, \dots, \lambda_m$, then $v_1, \dots, v_m$ is linearly independent.

**The idea**

Start with two. Could $v_2 = av_1$? Apply $T$ to both sides: $\lambda_2 v_2 = a\lambda_1 v_1$. Multiply both sides by $\lambda_2$ instead: $\lambda_2 v_2 = a\lambda_2 v_1$. The left sides are equal, so $a(\lambda_1 - \lambda_2)v_1 = 0$, so $a = 0$, so $v_2 = 0$, which an eigenvector is not.

The trick: *applying $T$ and multiplying by $\lambda_k$ do the same thing to $v_k$ but different things to the other eigenvectors.* Subtracting removes $v_k$ and leaves a relation among fewer vectors, with each coefficient multiplied by a nonzero number $\lambda_i - \lambda_k$. For the general case we want the shorter relation to be among independent vectors; so pick the *first* $v_k$ that depends on the earlier ones.

*Proof.* 1. Suppose the list is dependent. By (W2) there is a smallest $k$ with $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$. Because $k$ is smallest, $v_1, \dots, v_{k-1}$ is independent.

2. Write $v_k = a_1 v_1 + \cdots + a_{k-1} v_{k-1}$.

3. Apply $T$: $\lambda_k v_k = a_1 \lambda_1 v_1 + \cdots + a_{k-1} \lambda_{k-1} v_{k-1}$.

4. Multiply step 2 by $\lambda_k$: $\lambda_k v_k = a_1 \lambda_k v_1 + \cdots + a_{k-1} \lambda_k v_{k-1}$.

5. Subtract: $0 = a_1(\lambda_1 - \lambda_k)v_1 + \cdots + a_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1}$.

6. $v_1, \dots, v_{k-1}$ is independent, so each $a_i(\lambda_i - \lambda_k) = 0$. The eigenvalues are distinct, so each $a_i = 0$.

7. Then $v_k = 0$ by step 2, contradicting that eigenvectors are nonzero.

□

- **Prop. 55 / LADR 5.12.** An operator on a finite-dimensional $V$ has at most $\dim V$ distinct eigenvalues: eigenvectors for distinct eigenvalues are independent, and an independent list has at most $\dim V$ vectors (W1).
- Ex. 50: $\sin x, \sin 2x, \dots, \sin Nx$ are independent, because they are eigenvectors of $f \mapsto f''$ with the distinct eigenvalues $-1, -4, \dots, -N^2$.
- The shift $(a_1, a_2, \dots) \mapsto (a_2, a_3, \dots)$ on $\mathbb{R}^\infty$ has every real $\lambda$ as an eigenvalue, with eigenvector $(1, \lambda, \lambda^2, \dots)$.

## §2.4 Computing eigenvalues (Ch. 11.2)

**Proposition** (Prop. 56: eigenvalues of an operator are eigenvalues of its matrix)
Fix a basis of $V$. Then

$$Tv = \lambda v \iff \mathcal{M}(T)\mathcal{M}(v) = \lambda \mathcal{M}(v).$$

*Proof.* $\mathcal{M}(T)\mathcal{M}(v) = \mathcal{M}(Tv)$ (W5) and $\mathcal{M}(\lambda v) = \lambda \mathcal{M}(v)$; two vectors are equal exactly when their coordinate columns are equal. □

**The $2 \times 2$ characteristic polynomial.** For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$:

1. $\lambda$ is an eigenvalue $\iff$ $\lambda I - A$ is not invertible $\iff$ $\det(\lambda I - A) = 0$ (W4).

2. Apply the $2 \times 2$ formula to $tI - A$:

$$\det(tI - A) = (t - a)(t - d) - bc = t^2 - (a + d)t + (ad - bc).$$

   Call it $\chi_A(t)$.

3. So the eigenvalues of $A$ are the zeros of $\chi_A$ in $\mathbb{F}$. Their sum is $a + d$ and their product is $ad - bc$.

4. Eigenvectors for $\lambda$: solve $(A - \lambda I)x = 0$. The two rows are proportional, so use one: $(a - \lambda)x + by = 0$ gives $(b,\ \lambda - a)$; or $cx + (d - \lambda)y = 0$ gives $(\lambda - d,\ c)$. Use whichever is nonzero.

(Only the $2 \times 2$ formula is available for now; the characteristic polynomial in general comes later in the course.)

- Ex. 51: the rotation $R$ and the Fibonacci matrix $F$ have characteristic polynomials

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \chi_R = t^2 + 1; \qquad F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \chi_F = t^2 - t - 1.$$

  So $R$ has no real eigenvalues, and over $\mathbb{C}$ its eigenvalues are $\pm i$. The eigenvalues of $F$ are $(1 \pm \sqrt{5})/2$.

- **Ex. 52: use the structure, not a determinant.** $T(A) = A^{\mathsf{T}}$ on $\mathbb{C}^{2,2}$ satisfies $T^2 = I$. If $TA = \lambda A$ with $A \neq 0$, then $A = T^2 A = \lambda^2 A$, so $\lambda^2 = 1$. $E(1, T)$ is the symmetric matrices, spanned by $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$; $E(-1, T)$ is spanned by $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

## §2.5 PageRank (Ch. 11.3)

**The question.** Rank $n$ web pages by importance using only the links, where a page is important if important pages link to it.

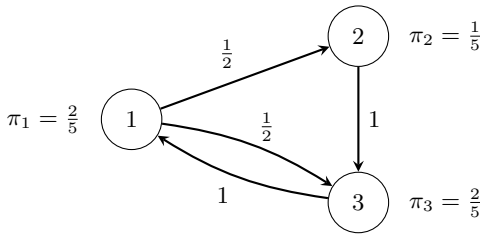
**Setting up.**

- Adjacency matrix: $C_{ij} = 1$ if page $i$ links to page $j$, else 0. $(C^k)_{ij}$ counts directed walks of length $k$ from $i$ to $j$.
- Let $d_j$ be the number of links out of page $j$ (assume $d_j > 0$). Define $P_{ij} = 1/d_j$ if page $j$ links to page $i$, else 0. Column $j$ sums to 1, so $P$ is stochastic. In matrix form $P = C^{\mathsf{T}} D^{-1}$ with $D = \operatorname{diag}(d_1, \ldots, d_n)$.
- A "random surfer" on page $j$ follows one of its links uniformly at random; if $p$ is the distribution of the surfer's position, $Pp$ is the distribution one step later.
- First attempt: rank by a stationary distribution $\pi = P\pi$. Coordinate $i$ says $\pi_i = \sum_j P_{ij}\pi_j$: the importance of page $i$ is collected from the pages linking to it.
- Problems: a page with no outgoing links (a **dangling page**) gives a zero column; the stationary distribution may not be unique; $P^k p_0$ may not converge. Dangling columns are replaced by the vector $v$ below.

Small example: pages $1, 2, 3$ with links $1 \to 2$, $1 \to 3$, $2 \to 3$, $3 \to 1$. Page 1 has two links, so each carries probability $\frac{1}{2}$; pages 2 and 3 have one link each.

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{pmatrix}.$$

Solving $P\pi = \pi$: $\pi_1 = \pi_3$, $\pi_2 = \frac{1}{2}\pi_1$, so $\pi = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$.




The link graph, with the probability on each link: column $j$ of $P$ lists the links out of page $j$. In the stationary distribution, what flows into each page equals its value: page 3 receives $\frac{1}{2} \cdot \frac{2}{5} + 1 \cdot \frac{1}{5} = \frac{2}{5}$.

**The PageRank equation.** Fix a **damping parameter** $0 < \alpha < 1$ and a probability vector $v$. At each step the surfer follows a link with probability $\alpha$, and with probability $1 - \alpha$ jumps to a page chosen according to $v$. The PageRank vector $r$ is the solution of

$$r = \alpha P r + (1 - \alpha)v, \qquad \text{equivalently} \qquad (I - \alpha P)r = (1 - \alpha)v.$$

**Proposition** (Prop. 57: eigenvalues of a stochastic matrix have absolute value at most 1)

**The idea**

A stochastic matrix redistributes mass without creating any. Measure the total mass of a (possibly complex) vector by $\|x\|_1 = |x_1| + \cdots + |x_n|$. Then $\|Px\|_1 \le \|x\|_1$. An eigenvector is scaled by $|\lambda|$, so $|\lambda| \le 1$.

*Proof.* 1. By the triangle inequality and $P_{ij} \ge 0$,

$$\|Px\|_1 = \sum_i \Big|\sum_j P_{ij} x_j\Big| \le \sum_i \sum_j P_{ij} |x_j|.$$

2. Exchange the sums:

$$\sum_j \Big(\sum_i P_{ij}\Big) |x_j| = \sum_j |x_j| = \|x\|_1,$$

because each column sums to 1.

3. If $Px = \lambda x$ with $x \neq 0$: $|\lambda| \, \|x\|_1 = \|Px\|_1 \le \|x\|_1$, and $\|x\|_1 > 0$, so $|\lambda| \le 1$.

□

**Theorem** (Prop. 58: existence and uniqueness of PageRank)

For a stochastic $P$, a probability vector $v$ and $0 < \alpha < 1$, there is exactly one $r \in \mathbb{R}^n$ with $r = \alpha P r + (1-\alpha)v$, and $r$ is a probability vector.

**The idea**

The equation is the linear system $(I - \alpha P)r = (1-\alpha)v$, so everything follows if $I - \alpha P$ is invertible. For a square matrix it is enough to show injectivity (W3). And $(I - \alpha P)x = 0$ says $Px = \frac{1}{\alpha}x$: that would make $\frac{1}{\alpha} > 1$ an eigenvalue of $P$, which Prop. 57 forbids. Uniqueness gives existence.

To see that $r$ is a probability vector, write the solution explicitly. Substituting the equation into itself:

$$r = (1-\alpha)v + \alpha P r = (1-\alpha)v + \alpha(1-\alpha)Pv + \alpha^2 P^2 r = \dots,$$

which suggests $r = (1-\alpha)\sum_{k \ge 0} \alpha^k P^k v$. This is a weighted average of the probability vectors $P^k v$ with weights $(1-\alpha)\alpha^k$ that sum to 1.

*Proof.* 1. *$I - \alpha P$ is injective.* If $(I - \alpha P)x = 0$ with $x \neq 0$, then $Px = \frac{1}{\alpha}x$, so $\frac{1}{\alpha} > 1$ is an eigenvalue of $P$, contradicting Prop. 57.

2. *Unique solution.* An injective operator on $\mathbb{R}^n$ is invertible (W3). So $(I - \alpha P)r = (1-\alpha)v$ has exactly one solution.

3. *The series.* Let

$$s = (1-\alpha) \sum_{k=0}^{\infty} \alpha^k P^k v.$$

Each $P^k v$ is a probability vector (W6), so each coordinate of each term lies in $[0, (1-\alpha)\alpha^k]$ and the series converges by comparison with a geometric series. The coordinates of $s$ are $\ge 0$.

4. *Sum of coordinates.*

$$\mathbf{1}^{\mathsf{T}} s = (1-\alpha) \sum_k \alpha^k \, \mathbf{1}^{\mathsf{T}} P^k v = (1-\alpha) \sum_k \alpha^k = 1.$$

So $s$ is a probability vector.

5. *$s$ solves the equation.*

$$\alpha P s + (1-\alpha)v = (1-\alpha) \sum_{k \ge 1} \alpha^k P^k v + (1-\alpha)v = s.$$

By uniqueness $s = r$.

□

**Computing $r$ by iteration.** Start from any probability vector $r_0$ and set $r_{k+1} = \alpha P r_k + (1-\alpha)v$.

1. Subtract $r = \alpha P r + (1-\alpha)v$: $r_{k+1} - r = \alpha P(r_k - r)$.
2. By step 1–2 of Prop. 57, $\|r_{k+1} - r\|_1 \le \alpha \|r_k - r\|_1$. So $\|r_k - r\|_1 \le \alpha^k \|r_0 - r\|_1 \to 0$: the error shrinks by a factor $\alpha$ each step.
3. The map $F(x) = \alpha P x + (1-\alpha)v$ satisfies $\|F(x) - F(y)\|_1 \le \alpha \|x - y\|_1$; such a map is a **contraction** (Banach fixed point theorem in analysis).

**The Google matrix.** Let $\mathbf{1} = (1, \dots, 1)^\mathsf{T}$ and $G = \alpha P + (1-\alpha)v\mathbf{1}^\mathsf{T}$.

- For a probability vector $x$, $\mathbf{1}^\mathsf{T} x = 1$, so $Gx = \alpha P x + (1-\alpha)v$. The PageRank equation is $Gr = r$: $r$ is an eigenvector of $G$ with eigenvalue 1.
- Let $H = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = 0\}$. $P$ keeps the sum of coordinates, so $H$ is invariant under $P$. For $x \in H$, $\mathbf{1}^\mathsf{T} x = 0$ and $Gx = \alpha P x \in H$. So $H$ is invariant under $G$, and on $H$ the matrix $G$ acts as $\alpha P$.
- $\mathbb{R}^n = \operatorname{span}(r) \oplus H$: $r \notin H$ since its coordinates sum to 1, and $\dim H = n - 1$. So $G$ fixes the direction $r$ and shrinks the invariant complement $H$ by at least the factor $\alpha$.
- History: Brin and Page used $\alpha = 0.85$ and uniform $v$; a non-uniform $v$ gives personalised PageRank.

**Shortcuts**

- $2 \times 2$: $\chi_A(t) = t^2 - (a+d)t + (ad - bc)$. Eigenvector for $\lambda$: $(b, \lambda - a)$ or $(\lambda - d, c)$.
- **Polynomial equation for $T$ gives the candidates**: if $Tv = \lambda v$ then $T^k v = \lambda^k v$, so $p(T)v = p(\lambda)v$. If $p(T) = 0$, every eigenvalue is a root of $p$. ($P^2 = P$: $\lambda \in \{0, 1\}$. $T^2 = I$: $\lambda = \pm 1$. $N^2 = 0$: $\lambda = 0$.)
- $n$ **distinct eigenvalues in dimension** $n$: the eigenvectors form a basis (Prop. 54 and counting). No further check needed.
- 0 **is an eigenvalue $\iff$ not invertible.**
- **Real matrix, complex eigenvalue**: if $Av = \lambda v$ then $A\bar{v} = \bar{\lambda}\bar{v}$ (take complex conjugates). Find one eigenvector and conjugate it.
- **Stochastic matrix**: 1 is always an eigenvalue. Reason: $P - I$ has columns summing to 0, so its range lies in $H$, so it is not surjective, so it is not injective (W3). If $P$ is also symmetric (rows sum to 1 too), the uniform vector is stationary.
- **Conjugate operators**: $S^{-1}TS$ has the same eigenvalues as $T$, with eigenspaces $S^{-1}E(\lambda, T)$ (HW6 P3).
- **Degree arguments on polynomial spaces**: differentiation lowers degree, so $p' = \lambda p$ forces $\lambda = 0$ (HW6 P2).
- **Invariant subspace check**: apply $T$ to a spanning list.

**Traps**

- An eigenvector is nonzero by definition; an eigenvalue may be 0.
- The sum of two eigenvectors with different eigenvalues is not an eigenvector. (Within one eigenspace, sums are fine.)
- Over $\mathbb{R}$ an operator may have no eigenvalue. In infinite dimensions an operator may have none even over $\mathbb{C}$.
- Independence needs *distinct* eigenvalues. Two eigenvectors with the same eigenvalue can be dependent.
- $\operatorname{null} T$ and $\operatorname{range} T$ are invariant, but an invariant subspace need not have an invariant complement: for $N$, the only invariant line is $\operatorname{span}(e_1)$.
- $|\lambda| \le 1$ for stochastic matrices is about absolute values: $\lambda = -1$ or complex $\lambda$ with $|\lambda| = 1$ can occur (permutation matrices).

# §3 Problem set

**HW6 P1.** Let $P\colon V \to V$ satisfy $P^2 = P$. (a) Prove that every eigenvalue of $P$ belongs to $\{0, 1\}$. (b) Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

*Hint:* (a) apply $P$ twice to an eigenvector. (b) $v = (v - Pv) + Pv$.

**HW6 P2.** Let $D\colon \mathcal{P}_n(\mathbb{F}) \to \mathcal{P}_n(\mathbb{F})$ be differentiation, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. (a) Determine all eigenvalues and eigenspaces of $D$. (b) Explain why the answer is different on the space of smooth functions on $\mathbb{R}$.

*Hint:* compare degrees in $p' = \lambda p$.

**HW6 P3.** Let $S\colon V \to V$ be invertible and $R = S^{-1}TS$. Prove that $T$ and $R$ have the same eigenvalues; more precisely $E(\lambda, R) = S^{-1}E(\lambda, T)$ for every $\lambda$.

*Hint:* rewrite $Rv = \lambda v$ as a statement about $Sv$.

**HW6 P4.** Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Using only the $2 \times 2$ determinant formula, prove $\det(tI - A) = t^2 - (a + d)t + (ad - bc)$. Then prove that $\lambda$ is an eigenvalue of $A$ if and only if it is a root of this polynomial.

**HW6 P5.** Find the eigenvalues and eigenspaces over the indicated field, and decide whether there is a basis of eigenvectors: $A_1 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (over $\mathbb{R}$, then over $\mathbb{C}$), $A_2 = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ over $\mathbb{R}$, $A_3 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ over $\mathbb{R}$.

**HW6 P6 (optional).** Let $T\colon V \to V$ and $v \neq 0$. Define $C(v, T) = \operatorname{span}(v, Tv, T^2v, \dots)$. (a) Prove that $C(v, T)$ is $T$-invariant. (b) If $V$ is finite-dimensional, prove there is $m \leq \dim V$ with $C(v, T) = \operatorname{span}(v, Tv, \dots, T^{m-1}v)$.

*Hint:* (b) take the first power $T^m v$ that depends on the earlier ones.

**HW6 P7 (optional).** For the stochastic matrix

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}$$

find a probability vector $p$ with $Pp = p$ and explain its meaning for the three-state Markov chain.

**R1 (Midterm 2 review, Problem 3).** For a polynomial $p$ and an operator $T$ with $Tv = \lambda v$, show $p(T)v = p(\lambda)v$.

**R2 (Midterm 2 review, Problem 7).** For any operator $T\colon V \to V$ prove that $\operatorname{null} T$ and $\operatorname{range} T$ are invariant under $T$, and that their annihilators are invariant under the dual map $T'$.

**R3 (Midterm 2 review, true/false).** (a) If $v_1, v_2$ are eigenvectors of $T$, then so is $v_1 + v_2$. (b) An operator on a real vector space always has an eigenvector.

**E1 (PageRank by hand).** For the three-page web in the text (links $1 \to 2$, $1 \to 3$, $2 \to 3$, $3 \to 1$) with $\alpha = \tfrac{1}{2}$ and uniform $v$, solve $r = \alpha P r + (1 - \alpha)v$.